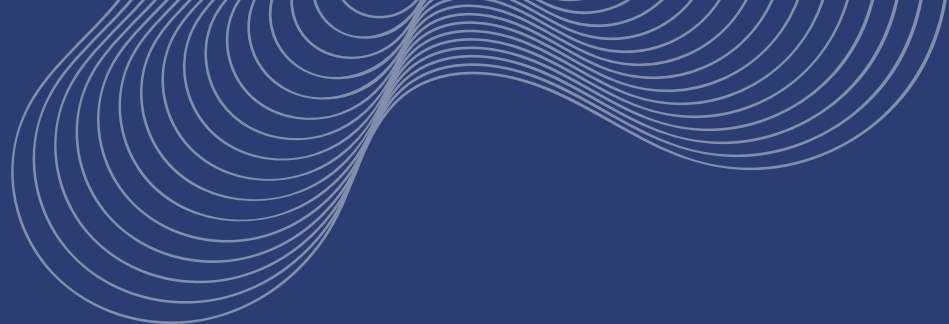




OptiGPU

PRESENTED BY : TEAM SD-014 (MAIT)

TEAM MEMBERS: 1) ISHAN MEHTA
2) HIMANSHI JAIN
3) ASHWIN BANSAL



CHALLENGES FACED

Cost Uncertainty – Hard to predict GPU costs for ML tasks

Performance Variability – GPU efficiency differs by model type

Regional Availability – Prices and availability change by region

Resource Matching – Complex to balance GPU, RAM & budget

Spot vs. On-Demand – Tough to trade off savings vs. stability

Task-Specific Needs–Training vs. inference need different setups

BUSINESS IMPACT

**Organizations overspend
by 30-45% on ML
infrastructure due to
suboptimal GPU instance
selection**



SOLUTION ARCHITECTURE

Frontend (React)

- Single- page application with responsive UI
- Real-time form validation
- Dynamic recommendation display
- Interactive instance comparison features



Backend (FastAPI)

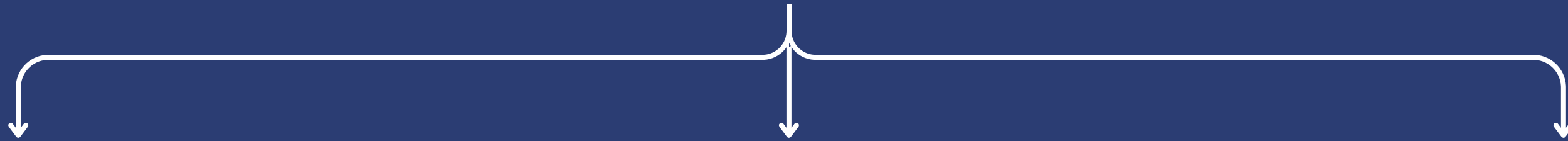
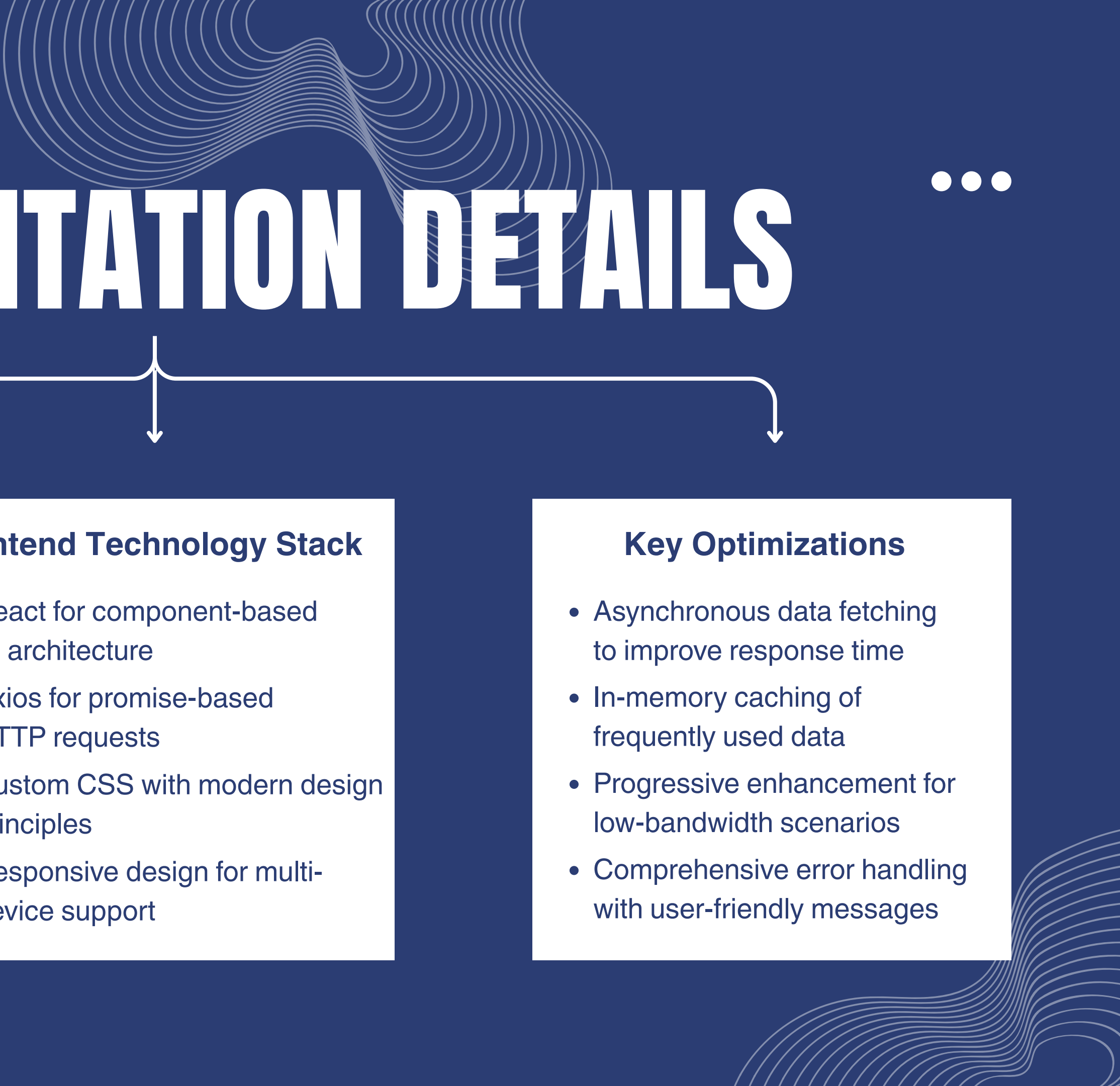
- RESTful API with async processing
- Multi-region data aggregation
- Intelligent scoring algorithm
- Fault-tolerant design with fallback options



Data Flow

- Users enter workload specs (task type, budget, etc.)
 - Backend fetches multi-region GPU options
 - Algorithm ranks instances based on fit
 - UI shows best matches with clear rationale
- 

IMPLEMENTATION DETAILS



Backend Technology Stack

- FastAPI for high-performance async API
- Pydantic for robust data validation
- AIOHTTP for parallel API requests
- - Python 3.8+ for modern language features

Frontend Technology Stack

- React for component-based UI architecture
- Axios for promise-based HTTP requests
- Custom CSS with modern design principles
- Responsive design for multi-device support

Key Optimizations

- Asynchronous data fetching to improve response time
- In-memory caching of frequently used data
- Progressive enhancement for low-bandwidth scenarios
- Comprehensive error handling with user-friendly messages

RESULTS & BENEFITS

For Data Scientists & ML Engineers

- 40% reduction in time spent selecting GPU instances
- 25-35% average cost savings on ML infrastructure
- Better workload-to-resource matching improving model performance

For DevOps & Infrastructure Teams

- Standardized approach to GPU provisioning
- Better cost forecasting and budget management
- Simplified infrastructure decision-making process

For Organizations

- Optimized cloud spending on ML workloads
- Faster time-to-market for ML projects
- Better resource utilization across regions

Future Enhancements

- Historical pricing analysis for trending recommendations
- ML workload profiling for more accurate matching
- Multi-cloud provider support beyond ACE Cloud
- Custom scoring weight adjustments for user preference

THANK YOU

Workload Requirements

Model Type

BERT

Dataset Size (GB)

9.8

Task Type

Training

Budget (USD/hour)

Optional

Preferred Region

Any Region

Get Recommendations

Recommended GPU Instances

gpu-v100-medium

Best Match

GPU: NVIDIA Tesla V100 (32GB)
vCPUs: 8
RAM: 32 GB
Region: ap-south-mum-1 (USA)

On-Demand: \$2.30/hr
Spot Price: \$0.69/hr Save 70%
Monthly: \$1679.00

Match Summary

gpu-a100-large

Option 2

GPU: NVIDIA A100 (80GB)
vCPUs: 16
RAM: 64 GB
Region: ap-south-mum-1 (USA)

On-Demand: \$4.50/hr
Spot Price: \$1.35/hr Save 70%
Monthly: \$3285.00

Match Summary

gpu-v100-medium

Option 3

GPU: NVIDIA Tesla V100 (32GB)
vCPUs: 8
RAM: 32 GB
Region: ap-south-noi-1 (USA)

On-Demand: \$2.30/hr
Spot Price: \$0.69/hr Save 70%
Monthly: \$1679.00

Match Summary